

TOMÁS S. R. SILVA

Curriculum Vitæ

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EDUCATION

PhD, Mathematics University of Campinas (UNICAMP) Supervisor: Prof. Henrique N. Sá Earp	Aug 2022 – Feb 2027 (expected) Campinas, São Paulo, Brazil
BSc, Computer Engineering University of Campinas (UNICAMP) Supervisor: Prof. Ricardo Dahab	Mar 2017 – Dec 2021 Campinas, São Paulo, Brazil

RESEARCH EXPERIENCE

Visiting Research Student City, University of London Supervisor: Prof. Yang-Hui He (London Institute for Mathematical Sciences & Merton College, Oxford)	Jan 2024 – Jul 2024 London, UK
PhD Researcher Laboratory of Gauge Theory and Algebraic Geometry (GTAG), UNICAMP	Aug 2022 – present
Junior Researcher Security and Cryptography Laboratory (LASCA), UNICAMP	May 2018 – Dec 2021

HONOURS & AWARDS

Academic Distinction Granted by the Institute of Computing, UNICAMP, for excellence during the undergraduate program.	2022
Best Undergraduate Research Project Granted by the Institute of Computing, UNICAMP, for the work “ <i>Characterizing Lattices for Cryptography</i> ”, recognised as a significant contribution to scientific and technological research in Computing that year.	2021

PUBLICATIONS

• Reviewed Articles	
Neural and numerical methods for G_2-structures on contact Calabi–Yau 7-manifolds E. Heyes, E. Hirst, H. N. Sá Earp & T. Silva <i>Physics Letters B</i> , Vol. 878, 2026. $\diamond$ arXiv:2602.12438	2026
Metaheuristic Generation of Brane Tilings Y.-H. He, V. Jejjala & T. Silva <i>Physics Letters B</i> , Vol. 862, 2025. $\diamond$ arXiv:2412.19313	2025
Machine learning topology of Calabi–Yau links T. Silva & H. N. Sá Earp <i>Proceeding Series of the Brazilian Society of Computational and Applied Mathematics</i> , Vol. 11, No. 1, 2025.	2025
Efficient isochronous fixed-weight sampling with applications to NTRU D. L. Gazzoni Filho, T. Silva & J. López <i>IACR Communications in Cryptology</i> , Vol. 1, No. 2, 2024.	2024
Machine-learning Sasakian and G_2 topology on contact Calabi–Yau 7-manifolds D. Aggarwal, Y.-H. He, E. Heyes, E. Hirst, H. N. Sá Earp & T. Silva <i>Physics Letters B</i> , Vol. 850, 2024. $\diamond$ arXiv:2310.03064	2024

• Preprints & Forthcoming

Minimising Willmore Energy via Neural Flow 2026
 E. Hirst, H. N. Sá Earp & T. Silva
[arXiv:2604.04321](#)

A semicontinuous relaxation of Saito's criterion and freeness as angular minimization 2026
T. Silva
[arXiv:2604.02995](#)

The exterior derivative and the mean value equality in $\mathbb{R}^n$ 2025
 D. Fadel, H. N. Sá Earp & T. Silva
[arXiv:2510.00999](#) — to appear in *São Paulo Journal of Mathematical Sciences*.

INVITED TALKS

Mathematics as a game: exploring freeness of line arrangements with machine learning 2026
 Mathematics Department Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP).

A data-driven view of exterior calculus 2026
 DANGER: Data, Numbers, and Geometry — Banff International Research Station (BIRS).

Machine Learning Ricci-Flat Metrics 2025
 São Paulo Meeting on Geometry, Institute of Mathematics and Statistics (IME), University of São Paulo (USP).

From Theory to Computation: Unraveling Hyperplane Arrangements 2025
 Geometry, Algebra and Combinatorics Seminar, UNICAMP.

Machine Learning Ricci-Flat Metrics 2025
 Geometry Seminar, Institute of Mathematical and Computer Sciences (ICMC), University of São Paulo (USP).

Machine Learning Riemannian Metrics with Special Holonomy 2025
 Machine Learning and Mathematics, Korea Institute for Advanced Study (KIAS).

Machine Learning Ricci-Flat Metrics 2025
 AI+Math Seminar, Brazilian Institute of Data Science (BIOS).

From Theory to Computation: Unravelling Hyperplane Arrangements 2024
 Seminar on Commutative Algebra and Algebraic Geometry, Federal University of Pernambuco (UFPE).

Learning Geometry and Topology of Algebraic Varieties 2023
 AI+Math Seminar, Brazilian Institute of Data Science (BIOS).

GRANTS & FELLOWSHIPS

Proposal Title: Geometry, Topology and Data Science
 Funding Agency: São Paulo Research Foundation (FAPESP)
 Role: Doctoral fellow | Principal Investigator: Prof. Henrique N. Sá Earp
 Coverage Period: Sep 2022 – present

Proposal Title: Post-Quantum Cryptography
 Funding Agency: Samsung R&D Institute Brazil
 Role: Undergraduate research grantee | Principal Investigators: Prof. Ricardo Dahab & Prof. Julio López
 Coverage Period: Nov 2020 – Nov 2021

Proposal Title: Characterizing Lattices and Codes for Cryptography
 Funding Agency: São Paulo Research Foundation (FAPESP)
 Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
 Coverage Period: Nov 2019 – Oct 2020

Proposal Title: Characterizing Lattices for Cryptography
 Funding Agency: São Paulo Research Foundation (FAPESP) & National Council for Scientific and Technological Development (CNPq)
 Role: Undergraduate research grantee | Principal Investigator: Prof. Ricardo Dahab
 Coverage Period: May 2018 – Apr 2019 (FAPESP) & Aug 2019 – Nov 2019 (CNPq)

TEACHING & ADVISING

Courses — Teaching Assistant (UNICAMP)

Linear Algebra Aug 2025 – Dec 2025 & Feb 2026 – Jul 2026
 Vector Calculus (Calculus II) Feb 2025 – Jul 2025 & Feb 2026 – Jul 2026
 Analytic Geometry Aug 2022 – Dec 2022 & Aug 2023 – Dec 2023

Undergraduate Student Co-Supervision

Bianca Bosnardo April 2026 – March 2027
 Undergrad research project: “*An introduction to hyperplane arrangements*”.

SERVICE & PROFESSIONAL DEVELOPMENT

• Service to the Discipline and Profession

Co-organiser — “AI meets Geometry” Workshop 2026
 XXII Escola de Geometria Diferencial (EGD). egd.ufpi.edu.br

• Service to the Community and Public

Volunteer Mathematics Tutor — “Cursinho Popular Zilda Arns” Jan 2020 – Jan 2021
 A non-profit preparatory school supporting underprivileged students in preparing for Brazilian university entrance exams.

• Professional Development — Workshops & Conferences Attended

Inaugural Workshop of the Brazilian Center for Geometry, UNICAMP 2026
 Vector Bundles and their Applications in Algebraic Geometry and
 Related Topics (Bandoleros), UNICAMP 2025
 Machine Learning and Mathematics, Korea Institute for Advanced Study 2025
 Mathematics and Machine Learning Program, Harvard CMSA 2024
 XLIII National Congress of Applied and Computational Mathematics,
 Federal University of Pernambuco 2024
 London Geometry and Machine Learning (LOGML), Imperial College London 2024
 Bridges Meeting on Special Geometries and Stability, Institut d’Études Scientifiques, Cargèse 2024
 Hybrid Conference on AI-Mathematics, Rio de Janeiro State University 2024
 MATH-AmSud School on Geometry: Group Actions, Symmetries, Moduli and Beyond,
 University of Chile 2023
 EnCoRi — Encounter on Codes, Lattices and Information Theory, UNICAMP 2023
 XV ALGA — Commutative Algebra and Algebraic Geometry, IMPA 2023
 XVI Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2021
 XV Theses, Dissertations and Scientific Undergraduate Research, Computing Institute, UNICAMP .. 2020
 I Journey of Undergraduate Research in Codes, Cryptography and Information Theory, UNICAMP .. 2019
 Metric and Combinatorial Problems Related to ECC, UNICAMP 2019
 XIV Theses, Dissertations and Undergraduate Research Workshop, Computing Institute, UNICAMP 2019
 XXVII Congress of Undergraduate Research, UNICAMP 2019
 I Latin American Week on Coding and Information, UNICAMP 2018

TECHNICAL SKILLS & LANGUAGES

Languages: Portuguese (native), English (advanced), Spanish (advanced), Hebrew (intermediate), French (intermediate).

Programming: C, C++, Python, Wolfram Mathematica, SageMath, Julia, SQL, Haskell, Linux/Bash, ARM Assembly.

Document Creation: L^AT_EX, Microsoft Office Suite, Markdown.